

Adaptive Worlds, Active Users: Physiology-Driven Adaptation During Active Interaction in Virtual Environments

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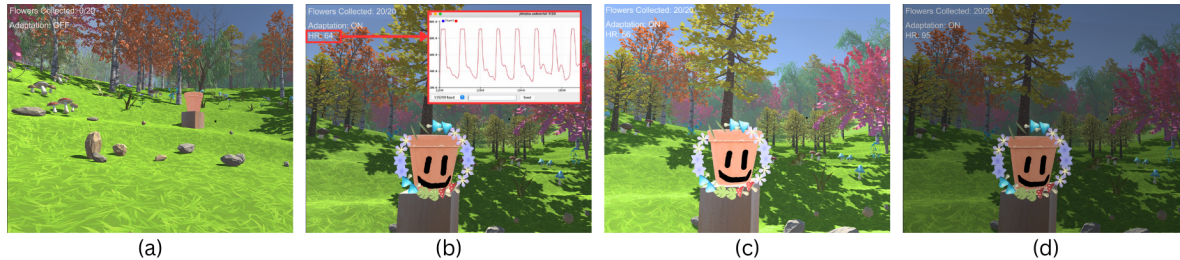


Figure 1: Teaser image illustrating real-time physiological adaptation in the interactive virtual garden. (a) Non-adaptive mode with static environmental conditions. (b) Adaptive mode enabled with heart rate at the defined baseline (70 BPM). (c) Low heart rate condition (below baseline by a threshold of 8 BPM) triggers increased scene brightness and visual liveliness. (d) High heart rate condition (above baseline by a threshold of 8 BPM) triggers dimmed lighting and reduced visual intensity as a part of adaptive ambient modulation.

ABSTRACT

We present a physiology-driven adaptive virtual environment that supports stress regulation during active interaction. The system integrates real-time heart rate sensing with a closed-loop adaptation pipeline to dynamically modulate environmental audio and lighting while users freely navigate and interact within an immersive virtual garden. Unlike many biofeedback-based VR experiences that emphasize passive viewing, this work demonstrates continuous physiological adaptation alongside object-based interaction, including drawing, object manipulation, and exploration. The adaptation logic operates in parallel with user activity, enabling uninterrupted engagement while maintaining responsive environmental modulation. This demo focuses on system design and technical validation, establishing a foundation for future studies on adaptive, interactive virtual environments for stress regulation.

Index Terms: Virtual Reality (VR), Adaptive Virtual Environments, Active Interaction, Stress Regulation

1 INTRODUCTION

Virtual reality (VR) has been widely explored for stress reduction and restorative experiences through immersive natural environments. However, many existing VR nature applications remain largely passive, offering limited interactivity and relying on pre-defined visual changes that do not respond to users' continuously changing physiological states [1, 3, 4].

Physiological signals such as heart rate provide an indicator of autonomic arousal and can be used to examine physiological fluctuations. While prior work has explored biofeedback-based VR, many systems emphasize passive viewing or isolated sensory adjustments.

We present a real-time adaptive virtual environment for an interactive virtual garden that integrates heart rate sensing with user-driven interaction to dynamically modulate environmental audio, lighting, and visual ambience. Rather than diagnosing or regulating stress, heart rate is collected as a physiological measure to compare adaptive and non-adaptive environments under different interaction conditions. This work focuses on system design and technical validation as a foundation for future experimental studies on adaptive, interactive VR.

2 SYSTEM OVERVIEW

The system combines real-time heart rate sensing with a closed-loop adaptation framework to regulate audiovisual ambience during active interaction in an immersive virtual garden.

2.1 Hardware and Sensing

The VR experience runs on a Meta Quest Pro head-mounted display, providing six-degree-of-freedom tracking and integrated spatial audio for immersive interaction. The application is developed using Unity.

Heart rate data is acquired using an Easy Pulse Sensor V1.1, connected to an Arduino Uno microcontroller. The PPG sensor continuously captures pulse signals from the user's fingertip, which are filtered and converted into beats-per-minute (BPM) estimates in real time.

The processed heart rate values are streamed to the VR application and used to support real-time environmental modulation while logging physiological responses for later comparison across conditions.

2.2 Adaptation Pipeline

As illustrated in Fig. 2, the system uses a closed-loop pipeline that maps real-time heart rate to environmental modulation by grouping physiological fluctuations into three internal system states.

Adaptation is applied through audio and lighting output channels. For each update cycle, the system adjusts ambient sound lay-

ering and global illumination intensity. Changes are applied gradually to preserve perceptual environmental coherence.

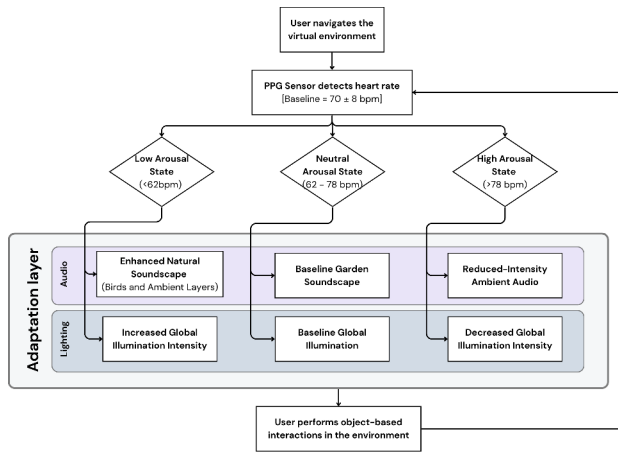


Figure 2: Closed-loop physiological adaptation pipeline for the interactive virtual garden. Real-time PPG-based heart rate sensing classifies user arousal states (low, neutral, high) to dynamically modulate environmental audio and lighting during continuous object-based interaction.

The adaptation loop runs in parallel with user-driven interaction, allowing physiological sensing and environmental modulation to proceed continuously during navigation and object-based actions. This enables active engagement without interrupting task flow.

The mapping between heart rate and environmental modulation was guided by prior biofeedback and affective computing literature, where physiological fluctuations are commonly mapped to gradual changes in sensory intensity. Audio and lighting were selected as primary output channels due to their strong perceptual influence and low interruption cost during active interaction. This implementation focuses on establishing a controllable baseline adaptation logic rather than optimizing affective or regulatory outcomes at this stage.

3 INTERACTION TYPES

Fig. 3 provides an overview of the primary user interaction modalities supported in the adaptive virtual garden, including pot drawing, rock manipulation, nature bracelet assembly, and free exploration. These interaction modes are distributed throughout the environment and operate alongside the real-time physiological adaptation pipeline.

Pot Drawing Interaction. Users are able to draw directly on virtual pots using controller-based input, enabling open-ended creative expression through freehand surface manipulation.

Rock Manipulation Interaction. Users can pick up, throw, and stack virtual rocks, supporting exploratory movement and simple physical play within the environment.

Nature Bracelet Interaction. Users may collect environmental elements such as flowers and assemble them into a virtual bracelet, reinforcing a sense of personal accumulation and connection with the surrounding environment.

Free exploration. In addition to object-based actions, users are free to navigate the virtual garden at their own pace without any prescribed task sequence, supporting open-ended spatial exploration and discovery.

These interaction modalities were selected based on prior work on restorative and attention-supporting activities, and were designed to encourage continuous, low-cognitive-load activity, allowing physiological adaptation to occur without requiring task interruption or user stillness [2].

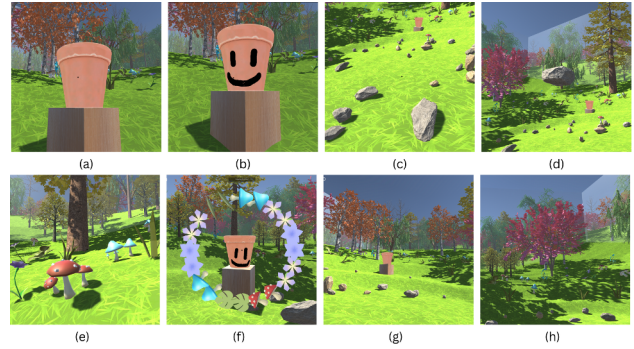


Figure 3: Overview of user interaction modalities in the adaptive virtual garden. (a–b) Pot interaction for freehand drawing and surface manipulation. (c–d) Rock interaction for object pickup, throwing, and stacking. (e–f) Nature bracelet interaction where users collect flowers and assemble wearable elements. (g–h) Free exploration and locomotion within the environment for spatial navigation and discovery. These interaction modes provide active user-driven input alongside real-time physiological adaptation.

4 TECHNICAL VALIDATION

The system operates in real time with continuous heart rate streaming and closed-loop audiovisual adaptation throughout user interaction. Physiological sensing, signal processing, and environmental modulation remain stable during extended runtime without observable data loss or perceptual latency. Adaptation updates occur smoothly and concurrently with object-based interaction and free navigation, preserving consistent system responsiveness during active use.

5 FUTURE WORKS

Future work will examine the relative contributions of different adaptation channels by comparing lighting-only, audio-only, and combined multi-modal modulation. Additional adaptive mechanisms, including dynamic object behavior, interaction difficulty, and spatial layout adjustments, will also be explored.

At present, the adaptation pipeline relies on a single physiological input channel, which may not fully capture the complexity of user states. Ongoing work will therefore examine the integration of additional input modalities, such as behavioral features, locomotion patterns, and interaction frequency, to improve the robustness of adaptive environmental modulation and its use as a controllable experimental factor. Controlled user studies will also be conducted to evaluate how these multi-input adaptive strategies influence user experience, engagement, presence, and physiological regulation within the virtual garden.

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